DIGITAL TWIN POC FOR PREDICTIVE HEALTHCARE

Happiest Health Reimagining & Reforming Healthcare in India Summit - 2026, Bengaluru

Project Title: Digital Twin PoC: Predictive Health Engine for Type 2 Diabetes & Cardiovascular Risk

Focus: Proactive Glycemic Spike Forecasting (2-Hours Ahead) & Continuous ASCVD Risk Scoring

Data Fusion: Static Indian EHR (Demographics, Conditions, Labs, Meds) + Dynamic Wearables (CGM, HR/HRV, Sleep, Steps)

Compliance: 100% Synthetic & Anonymized Pipeline compliant with India's DPDP Act and HIPAA

System Architecture & Data Flow

The system is engineered as an end-to-end pipeline connecting synthetic clinical sources to real-time AI inference and a doctor UI:

Layer	Components	Function / Role
Data Sources	Synthetic Indian EHR + Wearable IoT Generator	500 patients, CGM (5m), HR/HRV (1m), Sleep, Steps
Data Ingestion	EHRLoader, SensorLoader, StreamSimulator	Schema validation, time-series alignment, real-time generator
Processing Store	EHRPreprocessor, SensorPreprocessor, Fusion	Sliding window generation (72 steps), HRV extraction, normalization
Digital Twin Core	BiLSTM+Attention, XGBoost, Multi-Modal NN	2-hour glucose forecasting, ASCVD risk scoring, joint representations
Clinical Actions	PatientDigitalTwin, AlertSystem, What-If	Hysteresis debouncing, 4-tier alert triage, counterfactual simulation
Doctor Dashboard	Streamlit Clinician Portal (Port 8501)	Live telemetry, risk gauges, SHAP attribution, scenario simulator

Clinical Validation & Evaluation Metrics

All models were rigorously benchmarked on test data using clinically accepted standards:

Metric Category	Metric Name	Achieved Value	Clinical Standard / Target
Glucose Forecasting	Clarke Error Grid (Zones A+B)	**99.60%**	> 95.0% (Clinically safe)
Glucose Forecasting	MARD (Mean Abs Relative Diff)	**7.47%**	< 10.0% (Clinical grade CGM)
Glucose Forecasting	RMSE	**10.31 mg/dL**	< 25.0 mg/dL target
Glucose Forecasting	MAE	**8.36 mg/dL**	Low absolute error
Cardiovascular Risk	AUC-ROC	**0.965**	> 0.820 (State of the art)
Cardiovascular Risk	PR-AUC	**0.968**	High precision on imbalanced risk
Cardiovascular Risk	Sensitivity (Recall)	**89.80%**	> 85.0% (Minimize missed events)
Cardiovascular Risk	Specificity	**89.39%**	> 80.0% (Low false alarms)
Cardiovascular Risk	F1 Score	**0.8980**	High harmonic balance

Conclusion: The Digital Twin PoC fulfills all Phase 1 challenge criteria for Happiest Health Summit 2026, delivering high-precision multi-modal forecasting and clinician-actionable counterfactual simulations.